BAGEL · Frozen GEN + Spatial UND Velocity Correction

Method

❄ Frozen Trainable KV context

A BAGEL MoT · T2I

❄ UND expert

Prompt → embeddings

QKV

FFN

prompt KV

❄ GEN expert

vae2llm(x_t) + $T(t)$

QKV

FFN

image features · v_t

Multi-modal attention

× 28

B Training

1 Sample

x_0, ε, caption

$x_t = (1-t)x_0 + t\varepsilon$

2 GEN + CFG

v_{guided}

v_{cond}

$\hat{x}$

3 Target

$H_g(\varepsilon - x_0 - v_{\text{cond}})$

4 Fit

$r = \text{Corrector}(\hat{x})$

loss = MSE(r, target)

$g = v_{\text{guided}} - v_{\text{cond}}$ · $H_g(z) = z - \min(\langle z, g\rangle / \|g\|^2, 0)\, g$

❄ frozen: GEN · UND

trainable: corrector

C Inference

❄ prompt + image KV

GEN + CFG

v_{native}

Proposal

$\hat{x} = x_t - t\, v_{\text{native}}$

Corrector

r

Projection

$r_{\text{used}} = H_g(r)$

Euler

$v = v_{\text{native}} + r_{\text{used}}$

VAE → image

repeat $t: 1 \to 0$ · 50 nodes

r: velocity correction predicted by the corrector